Autistic Employment

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

https://cmu-intro-to-accessibility.github.io

Andrew Begel

Administrivia

- HW6 due Monday
- I'll be at the College Autism Summit next week
  - Monday: Patrick Carrington (in-person)
  - Wednesday: Franklin Li (Zoom)

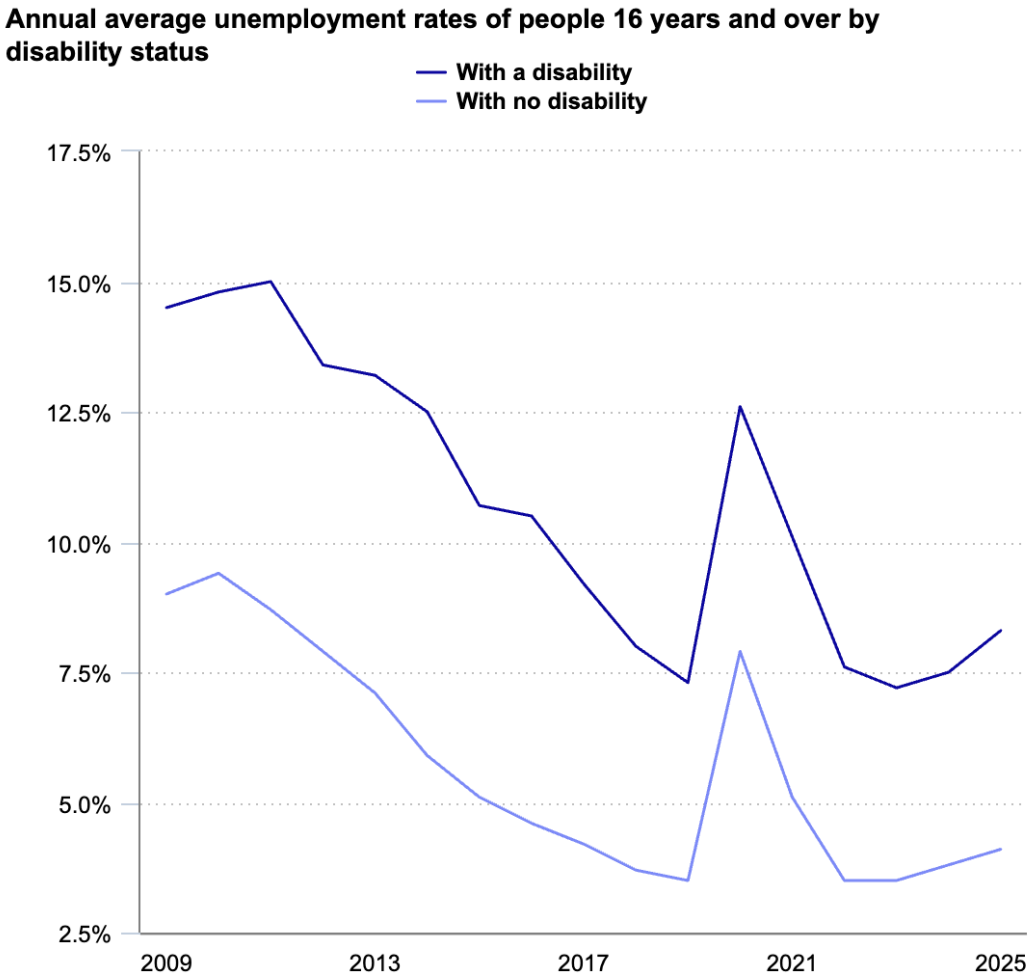
USA Unemployment Rate



Hover over chart to view data.

Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.
People whose ethnicity is identified as Hispanic or Latino may be of any race.

Disabled Unemployment Rate



Click legend items to change data display. Hover over chart to view data.
Annual estimates for 2025 are 11-month averages that exclude October. Data for October 2025 were not collected due to the federal government shutdown.
Source: U.S. Bureau of Labor Statistics.

The Disability Divide

The gap in societal inclusion for people with disabilities, including in education, employment, and access to technology.



Disability numbers are growing

- 20% of the world has a disability
- 15-20% of the population is neurodivergent.
- Mental health conditions are increasing



The employment divide is worsening

- 62% of people with disabilities in the USA are unemployed (vs 38% of non-disabled people).
- 50-85% of autistic people are unemployed!



Education gaps are increasing

- People with disabilities and the neurodivergent consistently have lower levels of education than their peers

What we need to do

“

For most people, technology makes things easier. For people with disabilities, technology makes things possible.

Mary Pat Radabaugh,
former Director of the IBM National Support Center for Persons with Disabilities

Focus areas:



Technology: affordable technology that is accessible by design



Talent development: skilling, education, and connection with job opportunities for people with disabilities



Workplace culture: fostering a welcoming and inclusive culture for people with disabilities

Software Engineering has a Talent Gap

Competitiveness in science, technology, and innovation requires taking full advantage of the domestic talent pool.

50 million Americans are *neurodivergent*, and they are vastly underutilized in the STEM workforce.



The Workplace is Often Inaccessible for Neurodivergent Individuals

They often suffer discrimination and stigma, preventing them from getting a job or staying employed.

This inaccessibility limits our capacity to leverage neurodivergent talents and create neuro-inclusive environments in the workplace.



Neurodivergent individuals bring many strengths to the workplace

Autism

- Attention to detail thinking
- Strong work ethic
- Work engagement
- Consistent performance on repetitive tasks
- Perseverance
- Honesty
- Pattern recognition
- Dependendability
- Out of the box

ADHD

- Creativity
- Hyper-focus
- Multi-tasking skills

However, Social Differences Bring Discrimination

Neurodivergent people have different styles of communication from non-autistic people.

This causes miscommunication and misunderstandings that hurt working relationships.

Many in power often prefer to work with those who think like they do and get rid of those who don’t.



Improving Working Relationships Will Reduce Discrimination

Software engineering roles require teamwork and effective communication and cooperation.

My research helps mixed-neurotype teams communicate and cooperate more effectively and willingly through better tools and education.





Andrew Begel
Associate Professor
CMU



Kathy Dow-Burger
Clinical Professor
U Maryland



Elizabeth Redcay
Professor
U Maryland



Jennifer Wessel
Professor
U Maryland



Steven Isaacson
Disability Strategist and LMSW
Sunflower Life Strategies

Become an Ally to Autistic Colleagues in the Workplace with INTENT

Discrimination & Employment Impact

- Sensory hyperacuity causes overload, reducing productivity and belonging
- Workplaces frequently fail to support sensory issues
- Co-workers misinterpret autistic coping strategies as incompetence or rudeness
  - Most non-autistic people lack knowledge about autism and only recognize autistic stereotypes.
- Autistic employees often get fired because “they don’t fit in.”
- Under- and unemployment rate for autistic adults is ~ 85%

Building Allyship: Awareness vs. Know-How

- Traditional workplace interventions
  - Focus on awareness or tolerance
  - Teach facts about autism
  - Aim to change knowledge or promote tolerance
  - Leave people unsure what to do in real situations

- Our approach: Teach allyship through INTENT
  - Allyship is learned behavior, not just belief
  - Teaches know-how: what to say, what to do, when to act
  - Centers practice in realistic workplace scenarios
  - Shifts responsibility to adjust workplace norms and environments

INTENT: Interactive Tool for Empathy in Neurotypicals

- Embodied as a transformational visual novel in the form of a web-based 3D video game.
- Designed to complement traditional information-based workplace training on allyship for autistic co-workers with experiential learning

Our Research Process

1

Interview autistic adults about workplace experiences

2

Prototype the game

3

Evaluate game with autistic adults

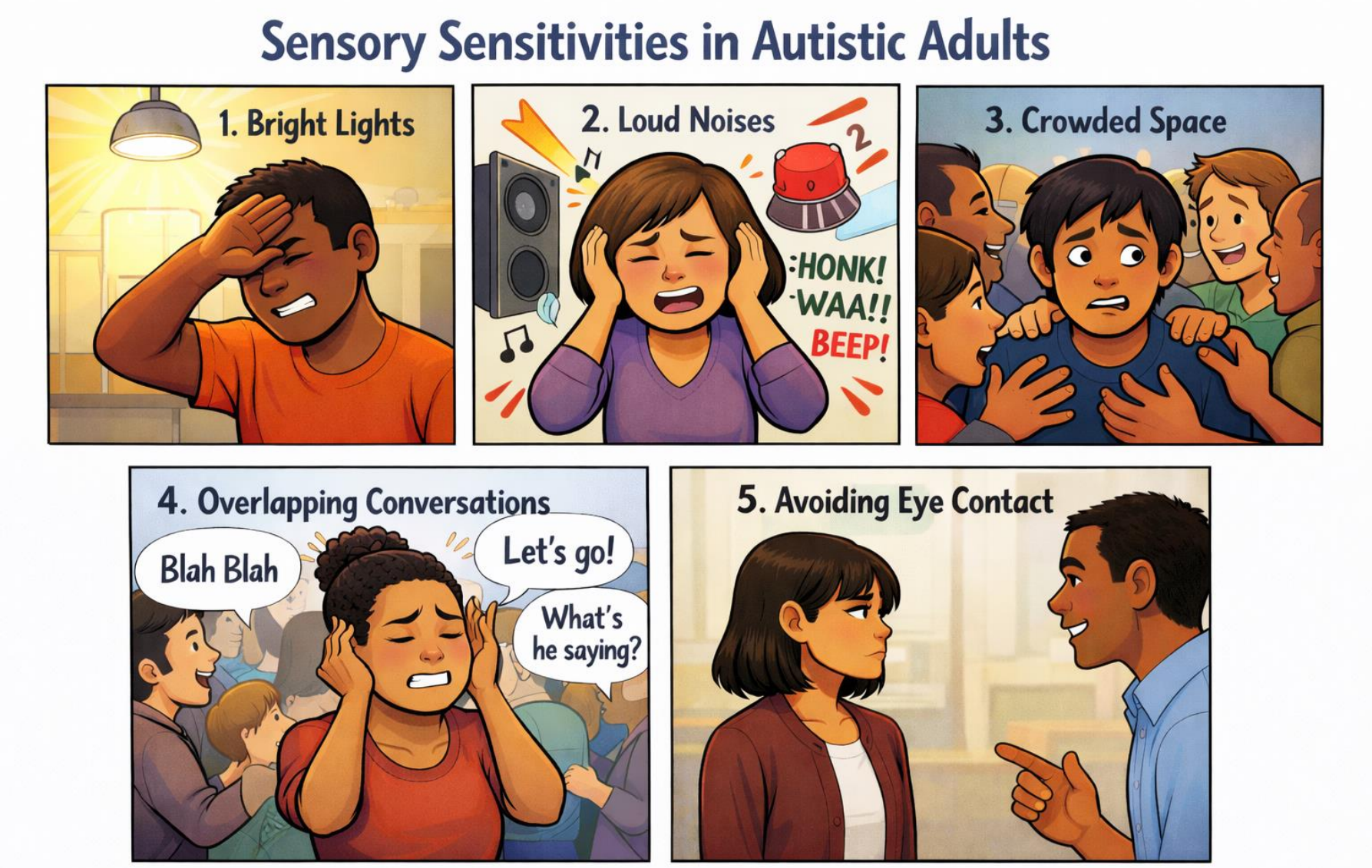
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Evaluate game with non-autistic adults

Studied autistic experiences of daily life

- Conducted 10 focus groups with 50 autistic adults
- Topics covered social experiences, noise/talking, light, eye contact, crowds
- Reflexive thematic analysis of 800 utterances produced 5 themes
  1. Sensory overstimulation
  2. Non-autistic reactions to overstimulated autistic people
  3. Coping strategies for overstimulation
  4. Self-perception of autistic strengths
  5. Socialization / Relaxation Behaviors

Theme 1: Sensory Overstimulation



Theme 1: Sensory Overstimulation

- “So, the more tired I am, the more I need like sunglasses or dark lighting or things like that because … it starts to give me headaches and I can’t concentrate. … It just gets really hard to handle.”
- “Abrupt sounds throw me off. It gives me difficulty with focusing on what I’m trying to do, or I might lose my train of thought.”
- “I do not like being touched, it makes me extremely uncomfortable.”
- “A lot of times when there are multiple conversations going on, it doesn’t matter where the sound is physically coming from, my head gives them all the same priority.”
- “I usually try to look at the bridge of someone’s nose, so it looks like I’m looking someone in the eye, even though I’m not.”

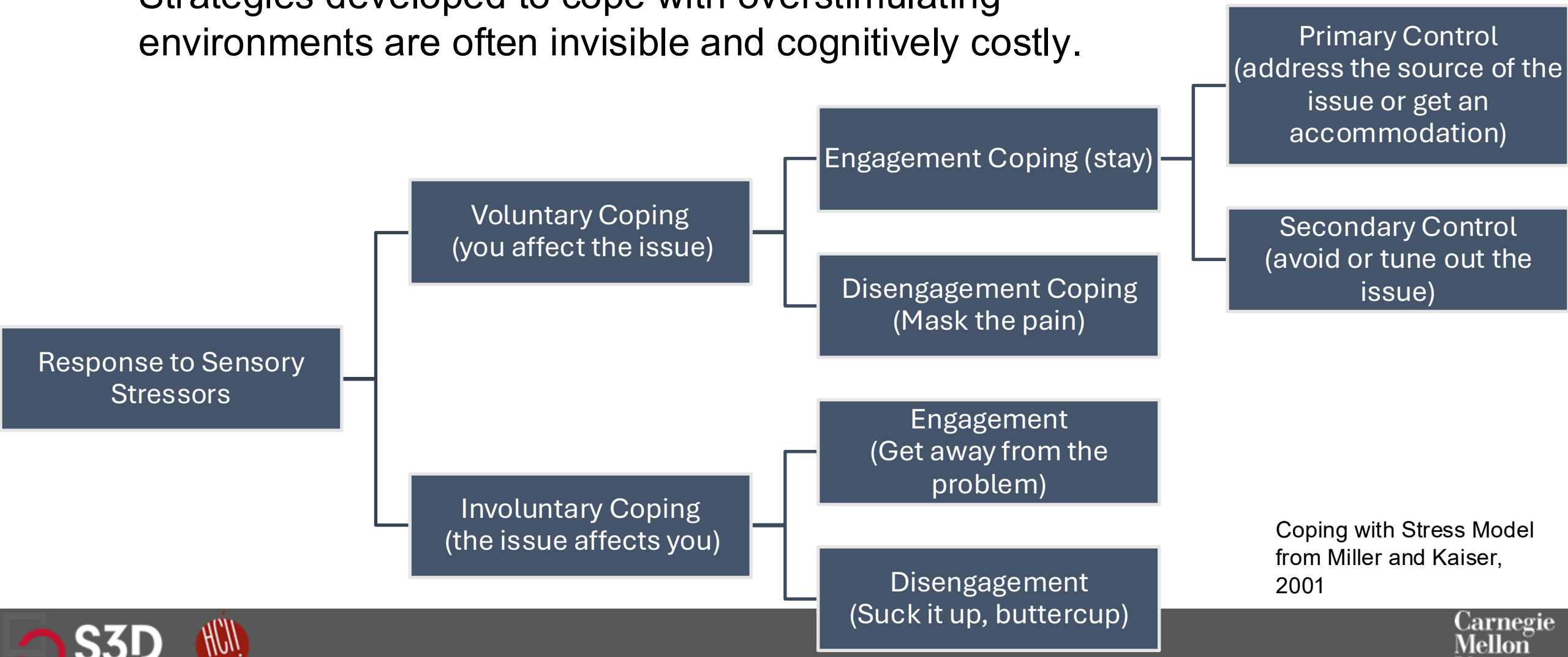
Theme 2: Non-autistic responses to overstimulated autistic people

- Responses range from supportive to mocking or even punitive

- “We should celebrate our differences instead of trying to mask them but we're so busy masking them to neurotypicals in order to survive.”
- “Sarcasm and humor is even worse sometimes because I take things very realistic. … Even my coworkers made fun of me because there was a guy who was super sarcastic and I’ll go ‘oh what really?' … It becomes a topic of teasing. It can be a little bit humiliating because you didn’t get it.”
- “Like at the workplace, sometimes people misread your facial looks, and if you’re not making eye contact, they’re thinking like you’re not listening to them … Those personal things [cause] a lot of stigma with autism.”

Theme 3: Coping Strategies for Overstimulation

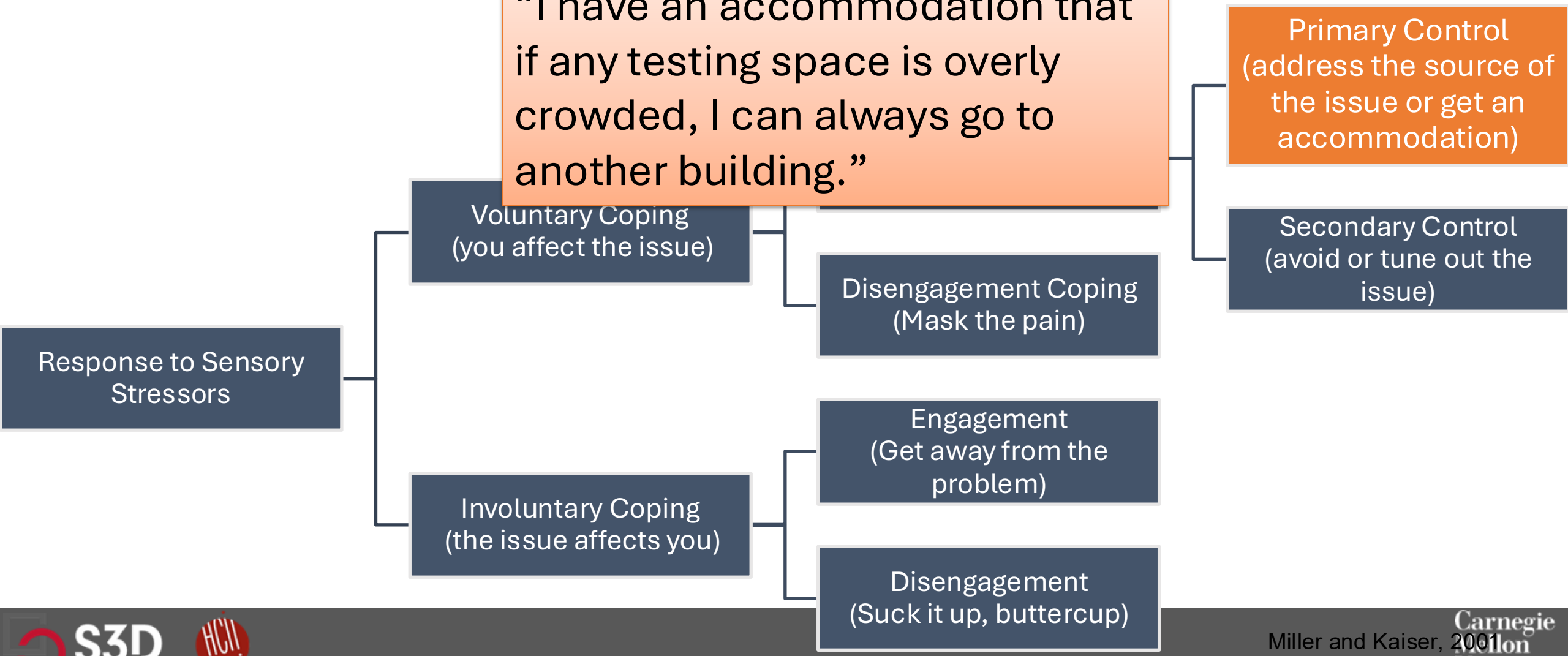
Strategies developed to cope with overstimulating environments are often invisible and cognitively costly.



Coping with Stress Model from Miller and Kaiser, 2001

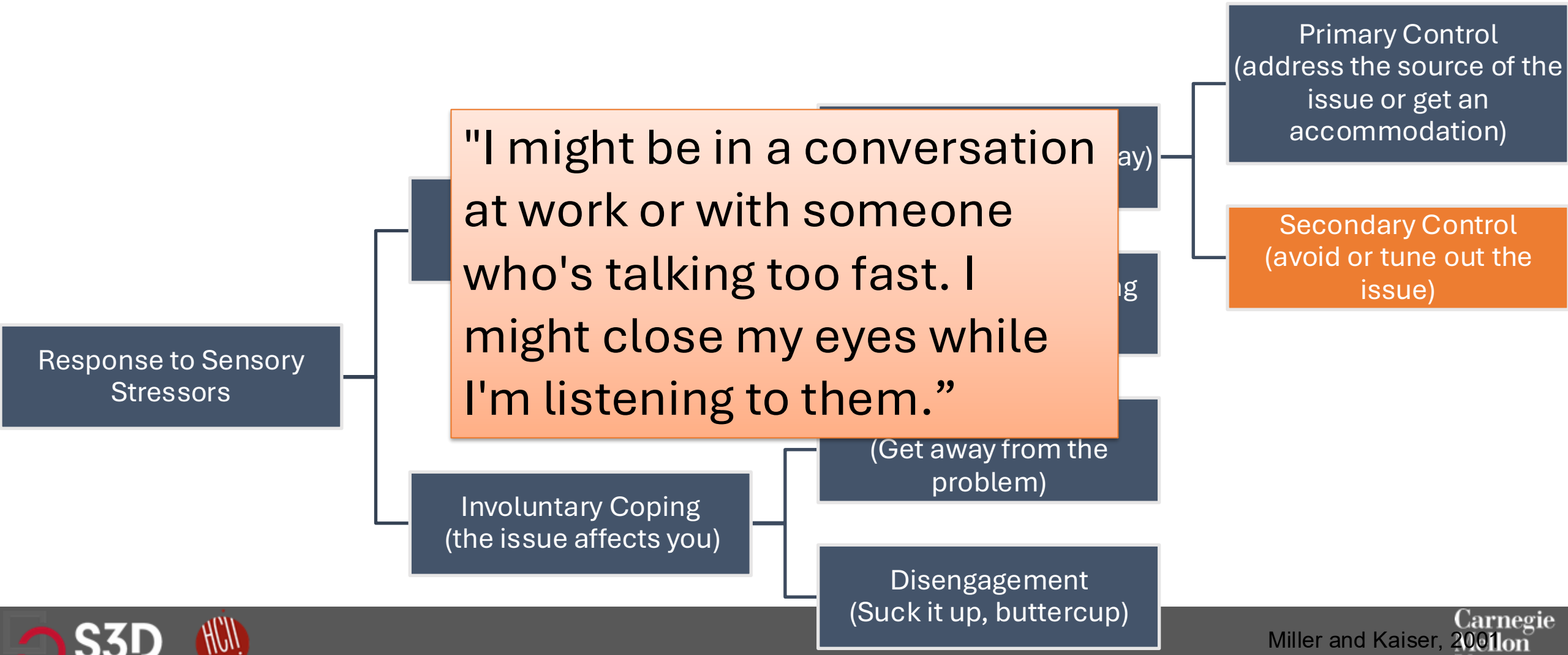
Theme 3: Coping Strategies for Overstimulation

"I have an accommodation that if any testing space is overly crowded, I can always go to another building."



Miller and Kaiser, 2001

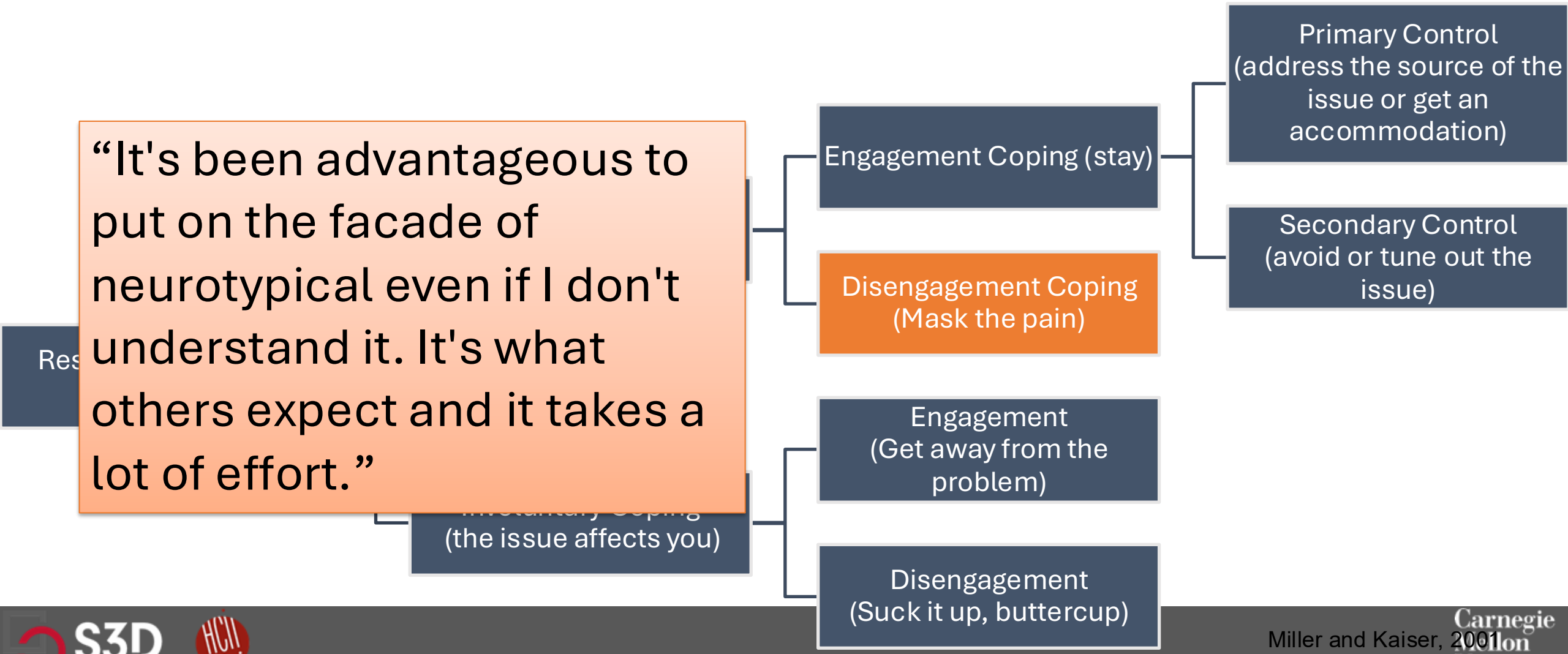
Theme 3: Coping Strategies for Overstimulation



"I might be in a conversation at work or with someone who's talking too fast. I might close my eyes while I'm listening to them."

Miller and Kaiser, 2001

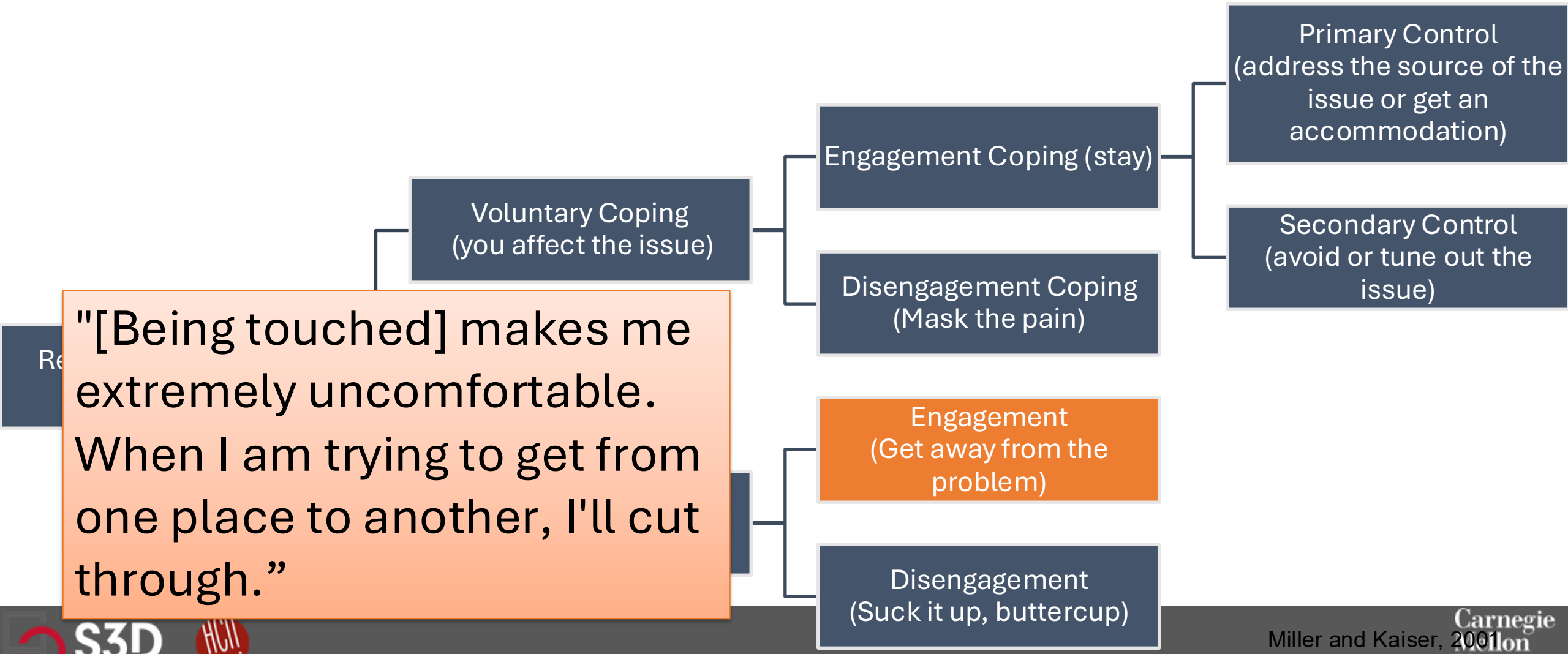
Theme 3: Coping Strategies for Overstimulation



"It's been advantageous to put on the facade of neurotypical even if I don't understand it. It's what others expect and it takes a lot of effort."

Miller and Kaiser, 2001

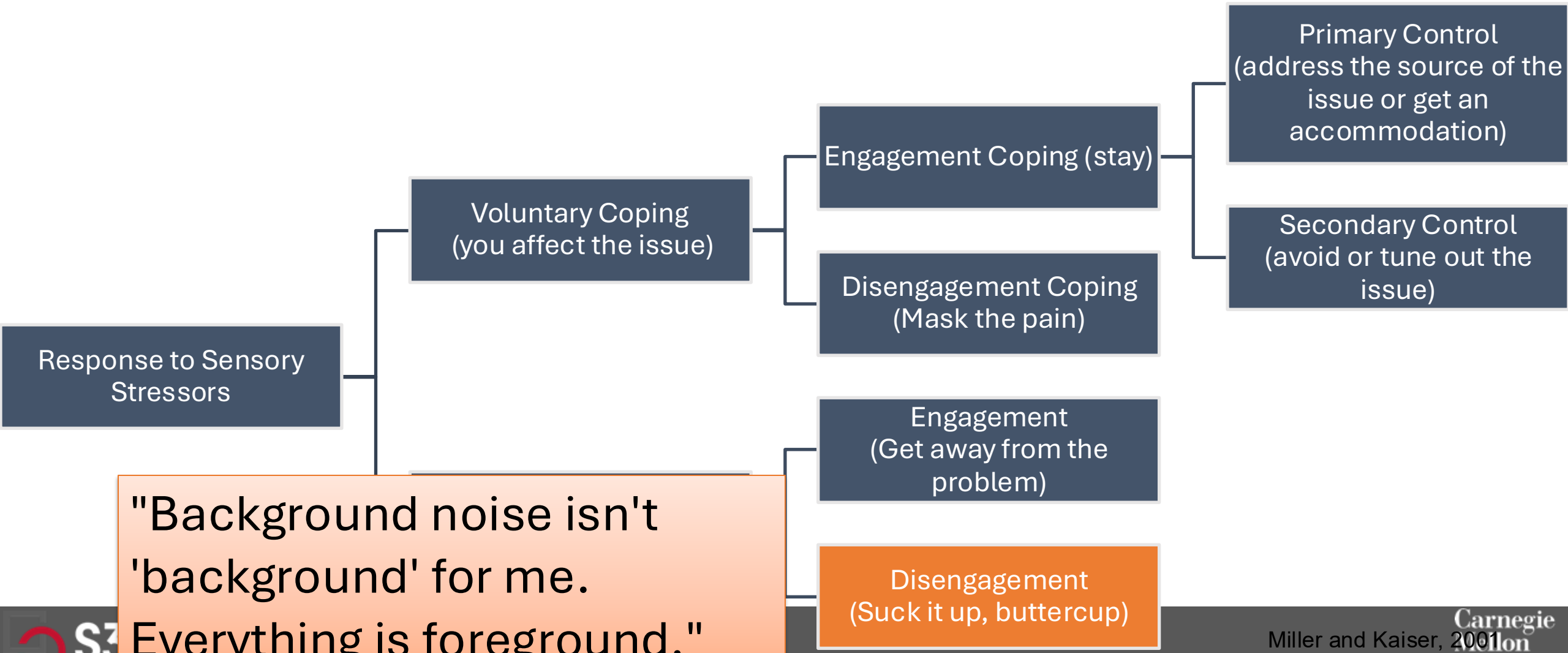
Theme 3: Coping Strategies for Overstimulation



"[Being touched] makes me extremely uncomfortable. When I am trying to get from one place to another, I'll cut through."

Miller and Kaiser, 2001

Theme 3: Coping Strategies for Overstimulation



"Background noise isn't 'background' for me. Everything is foreground."

Miller and Kaiser, 2001

Participation Activity

- How do you respond to stressful situations at school or work?
- Everyone count off 1, 2, 3, 4 (and repeat).
- Write one sentence about a stressful situation you experienced and a coping strategy you used that matches your number:
  1. Voluntary engagement (stay and do something about the problem)
  2. Voluntary disengagement (mask to pretend you don't care)
  3. Involuntary engagement (run away from the problem)
  4. Involuntary disengagement (suck it up and suffer)

Behavior Change through Workplace Training

- Autistic people have spent their whole lives learning how to fit into a non-autistic world.
- We flip the script with INTENT.
- INTENT trains non-autistic people to change their behaviors to accommodate their autistic colleagues and become their allies in the workplace.



Workplace Trainings Are Boring

- Can we augment informational lectures with more engaging training methods?
- We teach allyship by playing a game.
  - Games can help people understand and be more receptive to new information.

the Transformational Framework

A process tool for the development of Transformational games



The Transformational Framework is designed to help you create games that change players. The Framework is centered around eight exploratory questions critical for every team working in this space. Inside you'll find insights, best practices, and tips to help you navigate the challenges of developing transformational games.

www.transformationalframework.com

Sabrina Culyba



The Transformational Framework is based on the work of Schell Games, an independent game studio that specializes in bridging entertainment and education to create games and other experiences that are both engaging and enriching.

www.schellgames.com



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Focus on change over time, not one-off knowledge



Learn about autistic people

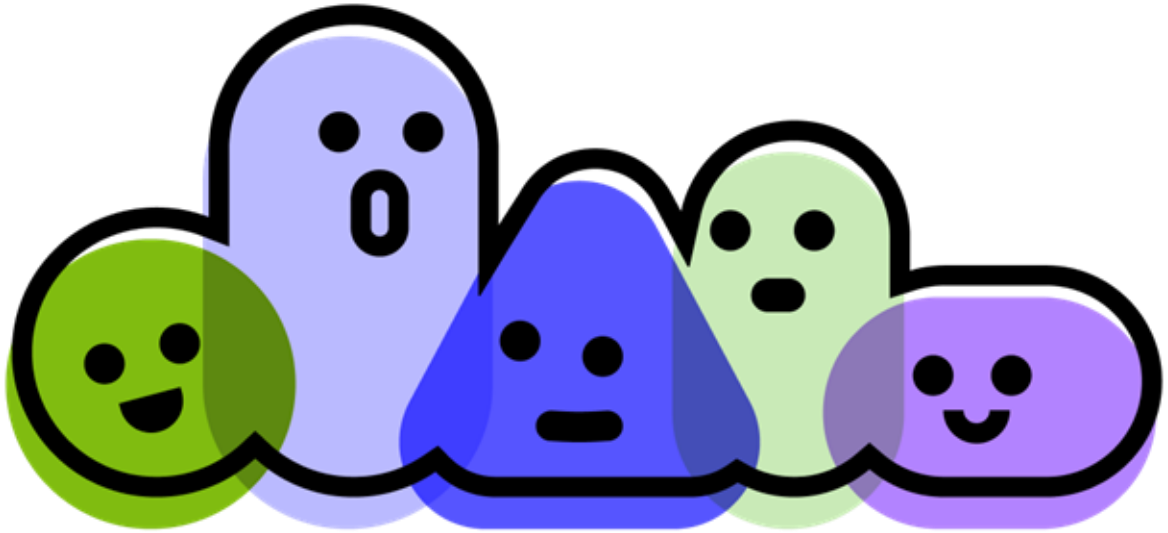
Grow empathy towards autistic people

Recognize discrimination
Intervene as an active bystander
Advocate for systemic change

Design Challenges

- Center autistic perspectives throughout design
- Represent heterogeneity of autistic people without having too many characters in the game

- Depict autistic people as multi-dimensional, not defined by deficits.
  - Avoid stereotypical appearances and dialogue
  - Emphasize the characters' intersectionalities.



INTENT

- We engaged with S3D PhD student Morgan Evans and ETC's capstone program to design and build the game.



Morgan Evans



Varun Girdhar | James Tseng | Zibo Ye | Ash Wang | Laura Yang

Game Mechanics

- Move your player character to the white circles to engage the narrative.
- Make choices about what to say or do (like a Choose Your Own Adventure book).
- See the consequences of your choices.
- After each scene, learn lessons by reflecting on critical moments.

Let’s Play Together!

https://cmu-variability.github.io/INTENT/

- Tips
  - Use Chrome.
  - Use a desktop or laptop
    - There’s a text input bug on touch screen devices
  - Turn off your speaker or click the Gear icon to turn off the sound effects and ambient sound.

What would you do?

- Scenario 1
  - Tony, a new employee, seems to misinterpret your jokes. Then, he inexplicably runs away from your presentation. Tony is acting strangely.
  - What do you do?
- Scenario 2
  - Ash is sensitive to bright lights and wants the blinds closed. Ming opens them anyway. Ming doesn't know that Ash is autistic.
  - How should you help Ash?
- Scenario 3
  - Tony is overwhelmed by an ill-defined project being forced upon him by your manager, Ali. How do you advocate for your friend's needs?
  - What should you say?

How did autistic people react to playing the game?

- Interview-based evaluation study
  - Run by Morgan Evans and MIII student Irene Choi
  - Augmented verbal interviews with Miro boards to enable less verbal participants to provide graphical feedback
- Research Questions
  1. How do autistic players relate to the autistic characters?
  2. How do autistic players believe non-autistic people will react to the game?
  3. What lessons do autistic players see in the game?
- Qualitative coding with reflexive thematic analysis with a constructivist approach

Findings: Resonance & Diversity

- Strong identification with autistic NPCs
  - “when Tony was shouting…when I get worked up I can shout too”
  - “I did personally feel that the noise was overwhelming at a certain point”
- Representation of autistic diversity
  - “I would say the game represented most aspects of the spectrum.”
  - [It’s a] “good thing…you had two autistic characters that are very diff[erent]”

Findings: Representation can be tricky

- Worry that non-autistic interpretations would rely on stereotypes
  - “[Non-autistic] people are gonna see [Tony] and be like, oh, yeah, all autistic people are like that.”
  - ”Not all people with autism necessarily have problems socializing.”
  - Characters became more complex throughout the game, diminishing stereotypes.

Findings: Lessons Conveyed

- Core lessons are valuable: Providing accommodations, advocating for autistic colleagues, demonstrating allyship

- Environmental lessons:
  - “mak[ing] the office quieter .. would benefit everyone.”
- Interpersonal lessons:
  - “glad that the character just stated their access needs”
- Structural lessons:
  - “The [third] scenario is centered around sticking up for Tony and his specific choice [that] it’s not a job for one person. … I see that it’s about being an ally. But, what I also see is the [more universal] need for context and details.”

Next Up: Pilot Testing in Companies

- Deliver allyship instruction to non-autistic workers in 2 conditions
  - Traditional lecture
  - Traditional lecture + INTENT game
- Evaluate how adding the game improves learning and alters implicit or explicit bias towards autistic people.